



Data Warehousing and Data Mining

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Reference :-

1. Data Mining Concepts and Techniques:

- Jiawei Han, Micheline Kamber, and Jian Pei;
Elsevier Science and Technology.

1. Data Mining and Data Warehousing Principles and Practical:

- Parteek Bhatia; Cambridge University Press.

2. Introduction to Data Mining:

- Pang-Ning Tan, Michael Steinbach and Vipin Kumar; Pearson Publication.



What is data mining?

- Data mining is also called *knowledge discovery from Databases* (KDD)
- Data mining is
 - extraction of useful patterns from data sources, e.g., databases, texts, web, images, etc.
- Patterns must be:
 - valid, novel, potentially useful, understandable



Classic data mining tasks

- **Regression:**
investigation of relationships between different sets of variables.
- **Classification:**
mining patterns that can classify future (new) data into known classes.
- **Association rule mining**
mining any rule of the form $X \rightarrow Y$, where X and Y are sets of data items. E.g.,
Cheese, Milk $\rightarrow$ Bread [sup =5%, confid=80%]
- **Clustering**
identifying a set of similarity groups in the data



Classic data mining tasks (contd)

- **Sequential pattern mining:**

A sequential rule: $A \rightarrow B$, says that event A will be immediately followed by event B with a certain confidence

- **Deviation detection:**

Discovering the most significant changes in data

- **Data visualization:**

Using graphical methods to show patterns in data.



Why is data mining important?

- Computerization of businesses produce huge amount of data
 - How to make best use of data?
 - Knowledge discovered from data can be used for competitive advantage.
- Online e-businesses are generate even larger data sets
 - Online retailers (e.g., amazon.com) are largely driving by data mining.
 - Web search engines are information retrieval (text mining) and data mining companies



Data-Rich with poor information

Huge amount of data, and no (powerful) data analysis tools, known as data rich but information poor situation !!!!!

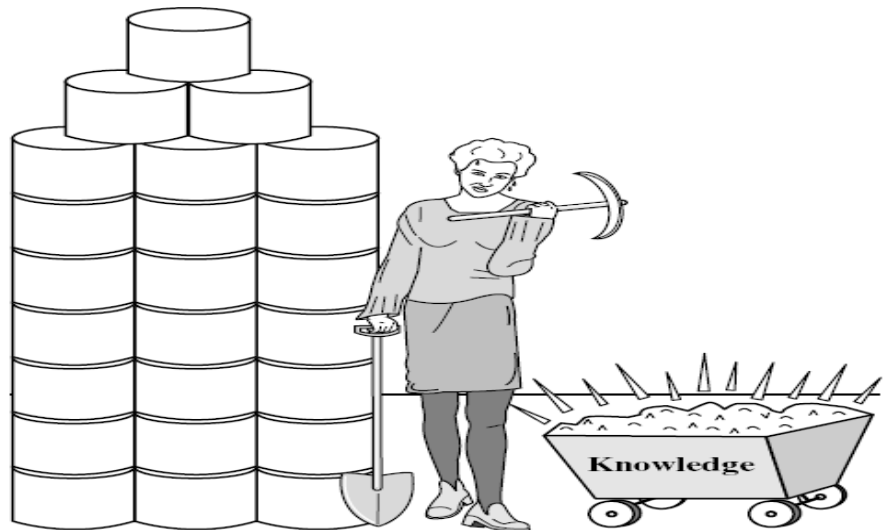




Data mining (searching for knowledge)

- Some people treat data mining as merely an essential step in the process of knowledge discovery.
- Data mining refers to extracting or mining knowledge from large amounts of data
 - Though, it is termed as data mining, it should have named as **“knowledge mining”**

So, data mining turns a huge collection of data into knowledge.





Why is data mining necessary?

- Make use of your data assets
- There is a big gap from stored data to knowledge; and the transition won't occur automatically.
- Many interesting things that one wants to find cannot be found using database queries
 - “find people likely to buy my products”
 - “Who are likely to respond to my promotion”
 - “Which movies should be recommended to each customer?”



Why data mining?

- The data is abundant.
- The computing power is not an issue.
- Data mining tools are available
- The competitive pressure is very strong.
 - Almost every company is doing (or has to do) it



Why data mining? Contd.

- The Explosive Growth of Data: from terabytes to petabytes (now a days Exabytes also)
 - Data collection and data availability
 - Automated data collection tools, database systems, Web, computerized society
 - Major sources of abundant data
 - Business: Web, e-commerce, transactions, stocks, ...
 - Science: Remote sensing, bioinformatics, scientific simulation,
 - Society and everyone: news, digital cameras, YouTube

We are drowning in data, but starving for knowledge!

- “Necessity is the mother of invention”—Data mining—Automated analysis of massive data sets



Related fields

- Data mining is an multi-disciplinary field:
 - Machine learning
 - Statistics Databases
 - Information retrieval
 - Visualization
 - Natural language processing
etc.



Data mining applications

- **Marketing:** customer profiling and retention, identifying potential customers, market segmentation.
- **Engineering:** identify causes of problems in products.
- **Scientific data analysis**, e.g., bioinformatics
- **Fraud detection:** identifying credit card fraud, intrusion detection.
- **Text and web:** a huge number of applications ...
- **Any application that involves a large amount of data ...**

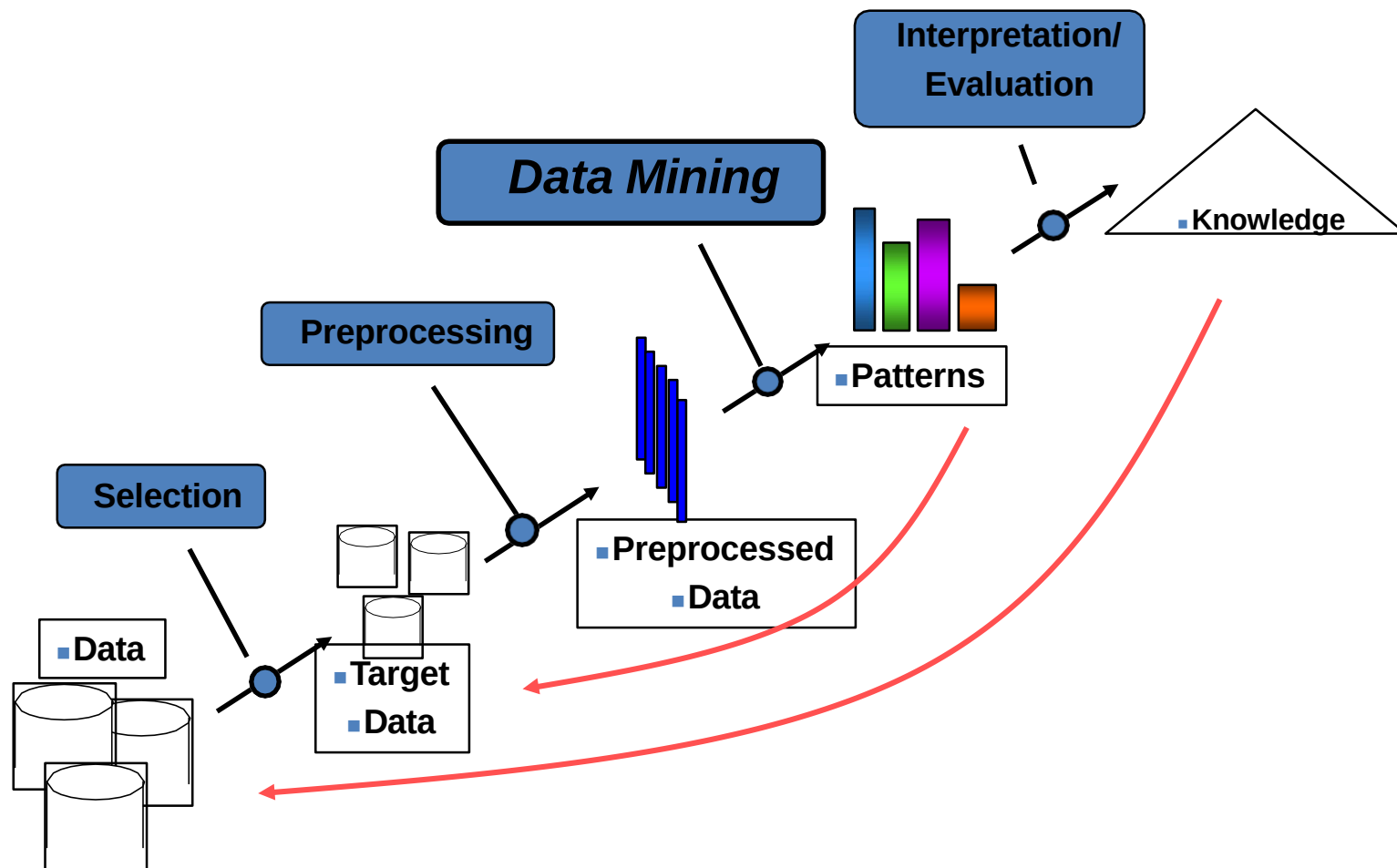


Components of data mining system

- **Database or data warehouse server:** responsible for fetching relevant data, based on the user's data mining request.
- **Knowledge base:** this is the domain knowledge that is used to guide the search or evaluate the interestingness of the resulting patterns.
- **Data mining engine:** consists of a set of functional modules for tasks such as characterization, association, correlation analysis, classification, prediction, cluster analysis, evolution analysis etc.
- **Pattern evolution module:** employs interesting measures and interacts with the data mining modules, to focus the search toward interesting patterns.
- **User interface:** this module communicates between the users and the data mining system.
 - Allowing the users to interact with the system by specifying a data mining query or task
 - Providing information to focus the search



KDD steps





Data mining (KDD) process

- Understand the application domain
- Identify data sources and select target data
- Pre-processing: cleaning, attribute selection, etc
- Data mining to extract patterns or models
- Post-processing: identifying interesting or useful patterns/knowledge
- Incorporate patterns/knowledge in real world tasks



Steps in knowledge discovery (1/2)

- **Data Cleaning:** where noisy and inconsistent data are removed
- **Data integration:** where multiple data source may be combined.
- **Data selection:** where data relevant to the analysis task are retrieved from the database.
- **Data transformation:** where data are transformed or consolidated into forms appropriate for mining, by performing summary or aggregation, etc.

These techniques may not be exclusive; they may work together, such as data cleaning can involve data transformation to correct wrong data.



Steps in knowledge discovery (2/2)

- **Data mining**: as essential process where intelligent methods are applied in order to extract data patterns.
- **Pattern evaluation**: identify the truly interesting patterns representing knowledge based on some interesting measures.
- **Knowledge presentation**: visualization and knowledge representation techniques are used to present the mined knowledge to the user.

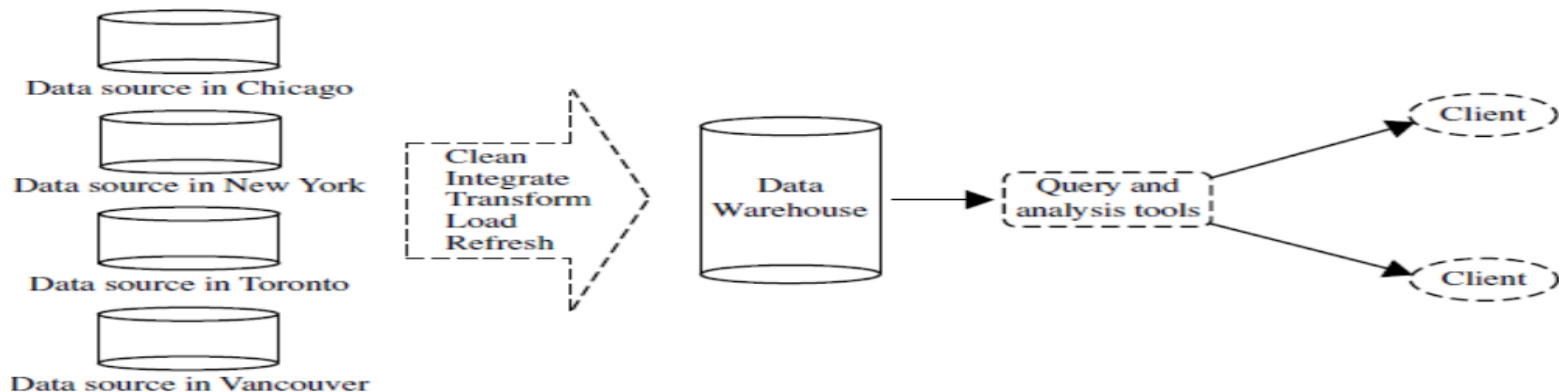
Data mining is the process of discovering interesting patterns and knowledge from large amounts of data. The data sources can include databases, data warehouses, the Web, other information repositories, or data that are streamed into the system dynamically.



What Kind of Data can be Mined?

Basic forms of data for mining applications are:

- Database data.
- Data warehouse data (will be taught in Data Warehouse section)



- Transactional data.

<i>trans_ID</i>	<i>list_of_item_IDs</i>
T100	I1, I3, I8, I16
T200	I2, I8
...	...

Data mining can also be applied to other forms of data.

- e.g., data streams, ordered/sequence data, graph or networked data, spatial data, text data, multimedia data, and the WWW.



Patterns Can Be Mined(1/7)

Descriptive: Descriptive mining tasks characterize properties of the data in a target data set.

Predictive: Predictive mining tasks perform induction on the current data in order to make predictions.

Data descriptions can be derived using:

- **Data characterization**, by summarizing the data of the class under study (often called the target class) in general terms
- **Data discrimination**, by comparison of the target class with one or a set of comparative classes (often called the contrasting classes),
- **Both data characterization and discrimination.**



Patterns Can Be Mined (2/7)

- **Data characterization** is a summarization of the general characteristics or features of a target class of data. The data corresponding to the user-specified class are typically collected by a query. Methods for data summarization and characterization are:
 - Statistical measures and plots
 - Data cube-based OLAP roll-up operation
 - Attribute-oriented induction technique
- **Data discrimination** is a comparison of the general features of the target class data objects against the general features of objects from one or multiple contrasting classes.
- The target and contrasting classes can be specified by a user, and the corresponding data objects can be retrieved through database queries. The methods used for data discrimination are similar to those used for data characterization.



Patterns Can Be Mined (3/7)

Mining Frequent Patterns, Associations, & Correlations

Mining Frequent Patterns: Examples

- **A frequent item set:** a set of items that often appear together in a transactional data set
 - for example, milk and bread, are frequently bought together in grocery stores by many customers.
- **A frequently occurring subsequence:** the pattern that customers, tend to purchase first a laptop, followed by a digital camera, and then a memory card, is a (frequent) sequential pattern.

A substructure can refer to different structural forms (e.g., graphs, trees, or lattices) that may be combined with item sets or subsequences. If a substructure occurs frequently, it is called a (frequent) structured pattern.



Patterns Can Be Mined(4/7)

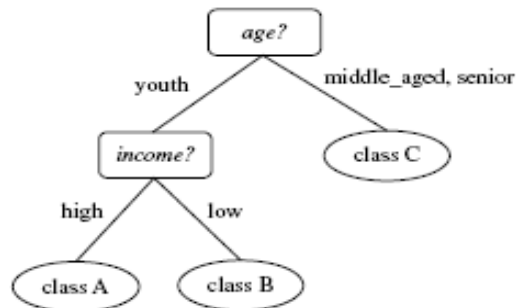
Classification and Regression for Predictive Analysis

- **Classification:** process of finding a model /function that describes and distinguishes data classes or concepts.
- The model - derived based on a set of **training data** (i.e., class labels are known). - predict the class label of objects (class label is unknown).

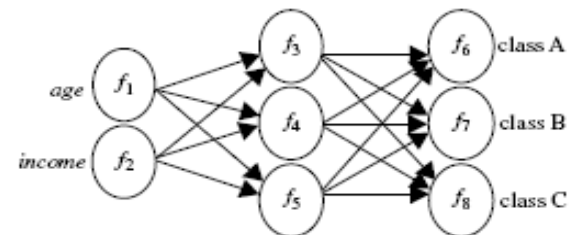
Various forms of representation of derived model are:

$age(X, \text{"youth"}) \text{ AND } income(X, \text{"high"}) \longrightarrow class(X, \text{"A"})$
 $age(X, \text{"youth"}) \text{ AND } income(X, \text{"low"}) \longrightarrow class(X, \text{"B"})$
 $age(X, \text{"middle_aged"}) \longrightarrow class(X, \text{"C"})$
 $age(X, \text{"senior"}) \longrightarrow class(X, \text{"C"})$

Classification model using IF-THEN



Classification model using Decision Tree



Classification model using Neural Network



Patterns Can Be Mined(5/7)

Classification and Regression for Predictive Analysis

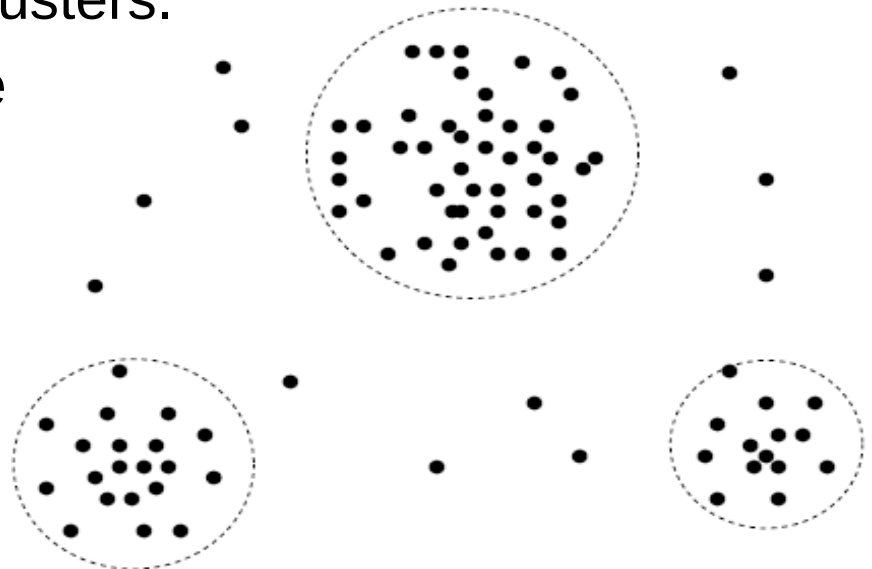
- **Regression** is used to predict missing or unavailable numerical data values rather than (discrete) class labels.
- The term prediction refers to both numeric prediction and class label prediction.
- **Regression analysis** is a statistical methodology that is most often used for numeric prediction, although other methods exist as well.
- Regression also encompasses the identification of distribution trends based on the available data.



Patterns Can Be Mined (6/7)

Cluster and Outlier Analysis

- Objects are clustered or grouped based on the principle of *maximizing the intra class similarity and minimizing the interclass similarity*.
- Clusters of objects are formed so that objects within a cluster have high similarity in comparison to one another, but are rather dissimilar to objects in other clusters.
- Each cluster so formed can be viewed as a class of objects, from which rules can be derived.





Patterns Can Be Mined(7/7)

Cluster and Outlier Analysis

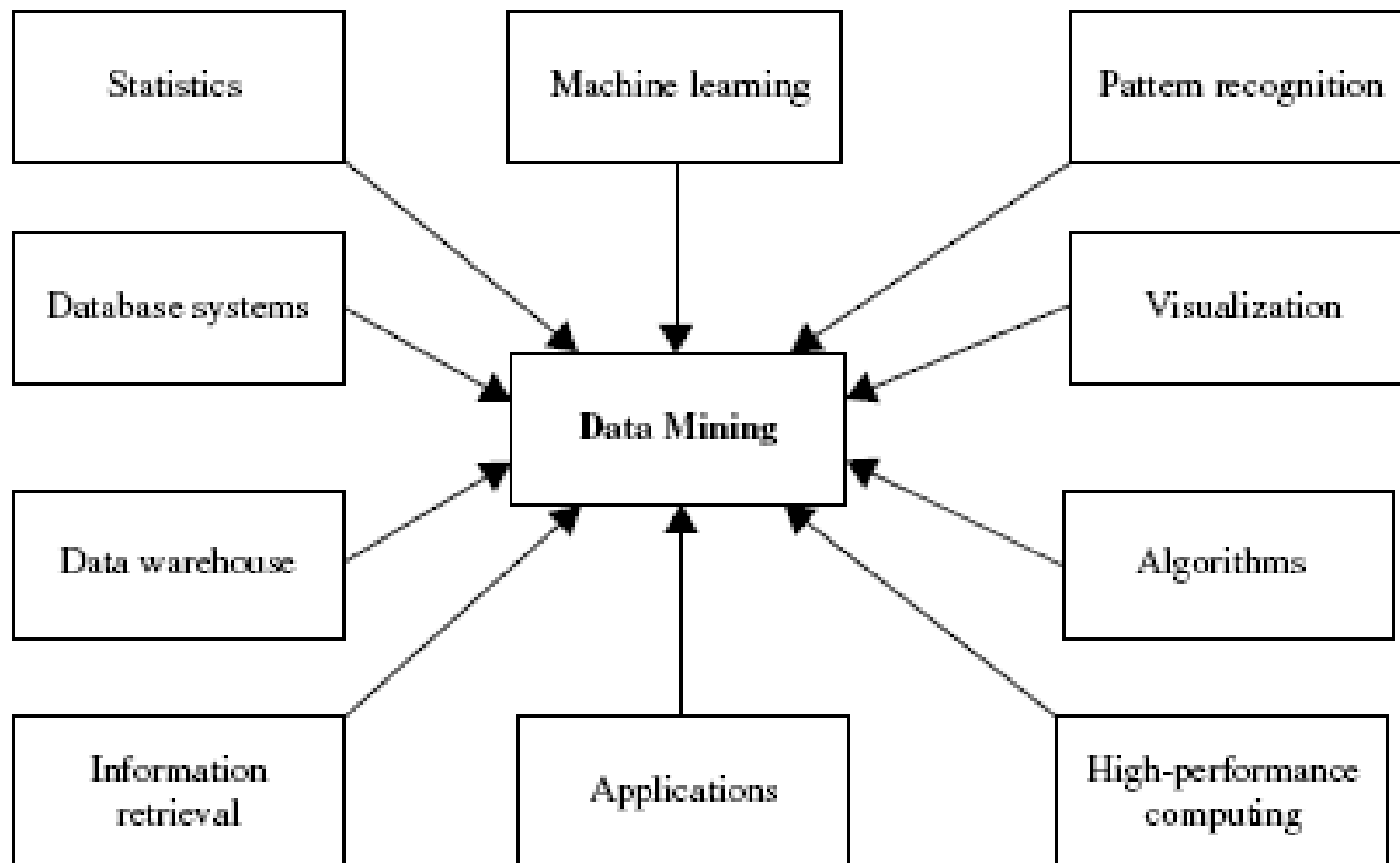
- A data set may contain objects that do not comply with the general behavior or model of the data. These data objects are outliers.
- Many data mining methods discard outliers as noise or exceptions. However, in some applications (e.g., fraud detection) the rare events can be more interesting than regularly ones.
- The analysis of outlier data is referred to as **outlier analysis** or **anomaly mining**.

Example: Outlier analysis may uncover fraudulent usage of credit cards by detecting purchases of unusually large amounts for a given account number in comparison to regular charges incurred by the same account.



Which Technologies Are Used?

Techniques from other domains incorporated in Data Mining are:





Major Issues in Data Mining?

Mining Methodology:

- Mining various and new kinds of knowledge,
- Mining knowledge in multidimensional space
- Data mining—an interdisciplinary effort
- Boosting the power of discovery in a networked environment
- Handling uncertainty, noise, or incompleteness
- Pattern evaluation and pattern- or constraint-guided mining

User Interaction:

- Interactive mining
- Incorporation of background knowledge
- Ad hoc data mining and data mining query languages
- Presentation and visualization of data mining results

Efficiency and Scalability:

- Efficiency and scalability of data mining algorithms
- Parallel, distributed, and incremental mining algorithms

Diversity of Database Types:

- Handling complex types of data
- Mining dynamic, networked, and global data repositories

Data Mining and Society:

- Social impacts of data mining
- Privacy-preserving data mining
- Invisible data mining



Multi-Dimensional Measure of Data Quality

- A well-accepted multidimensional view:
 - Accuracy
 - Completeness
 - Consistency
 - Timeliness
 - Believability
 - Value added
 - Interpretability
 - Accessibility



Why Data Preprocessing?

Data in the real world is dirty. How?

Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data

- e.g., occupation=" " (example of lacking attribute value)

Noisy: containing errors or outliers

- e.g., Salary="-10"

Inconsistent: containing discrepancies in codes or names

- e.g., Age="42" Birthday="03/07/1997"
- e.g., Was rating "1,2,3", now rating "A, B, C"
- e.g., Discrepancy between duplicate records



Why Data is dirty?

- **Incomplete data may come from**
 - “Not applicable” data value when collected
 - Different considerations between the time when the data was collected and when it is analyzed
 - Human/hardware/software problems
- **Noisy data (incorrect values) may come from**
 - Faulty data collection instruments
 - Human or computer error at data entry
 - Errors in data transmission
- **Inconsistent data may come from**
 - Different data sources
 - Functional dependency violation (e.g., modify some linked data)

N.B.: Duplicate records also need data cleaning



Why is Data Preprocessing important?

- **No quality data, no quality mining results!**
 - Quality decisions must be based on quality data
 - e.g., duplicate or missing data may cause incorrect or even misleading statistics.
 - Data warehouse needs consistent integration of quality data
- Data **extraction**, **cleaning**, and **transformation** comprise the majority of the work of building a data warehouse



Why is Data Preprocessing important?

- Real world databases are highly susceptible to noisy, missing, inconsistent due huge size and origin from multiple heterogeneous sources.
- Low-quality data will result low-quality mining result.
- How data can be preprocessed to improve the quality of the data, and consequently, that of mining results?
- How to preprocess the data so as to improve the efficiency of the mining process?



Major Tasks in Data Preprocessing

- **Data cleaning**
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data integration**
 - Integration of multiple databases, data cubes, or files
- **Data transformation**
 - Normalization and aggregation
- **Data reduction**
 - Obtains reduced representation in volume but produces the same or similar analytical results
- **Data discretization**
 - Part of data reduction but with particular importance, especially for numerical data



Descriptive Characteristics

- **Motivation**
 - To better understand the data: central tendency, variation and spread
- **Data dispersion characteristics**
 - median, max, min, quantile, outliers, variance, etc.
- **Numerical dimensions correspond to sorted intervals**
 - Data dispersion: analyzed with multiple granularities of precision
 - Boxplot or quantile analysis on sorted intervals
- **Dispersion analysis on computed measures**
 - Folding measures into numerical dimensions
 - Boxplot or quantile analysis on the transformed cube



Measuring the Central Tendency

- **Mean (algebraic measure) (sample vs. population):**

- Weighted arithmetic mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\mu = \frac{\sum x}{N}$
- Trimmed mean: chopping extreme values

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

- **Median:** A holistic measure

- Middle value if odd number of values, or average of the middle two values otherwise
- Estimated by interpolation (for *grouped data*):

- **Mode:**

$$median = L_1 + \left(\frac{n/2 - (\sum f)l}{f_{median}} \right) c$$

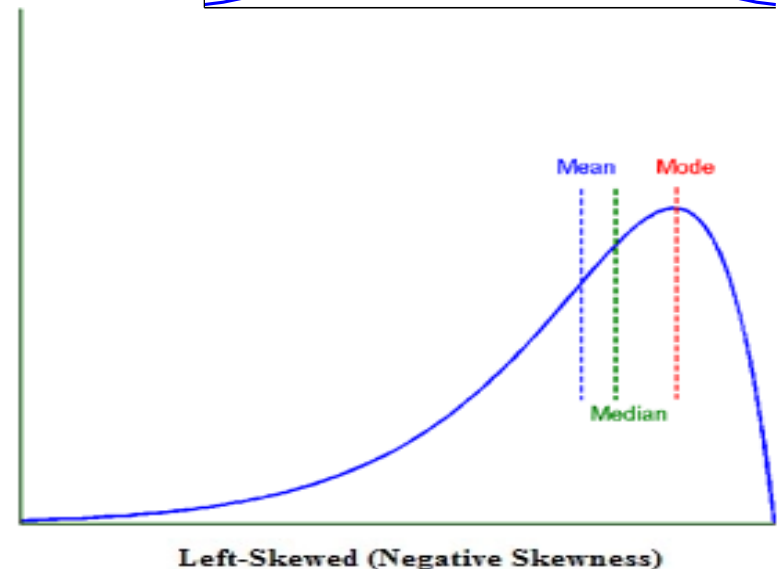
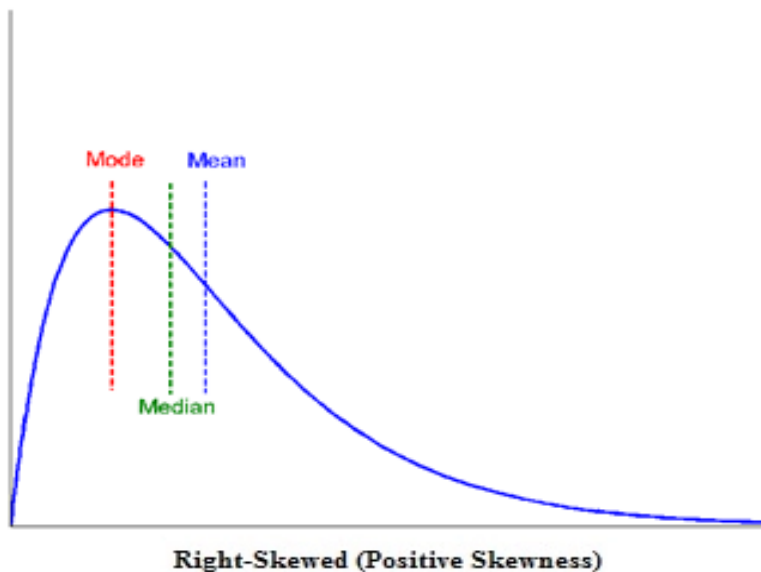
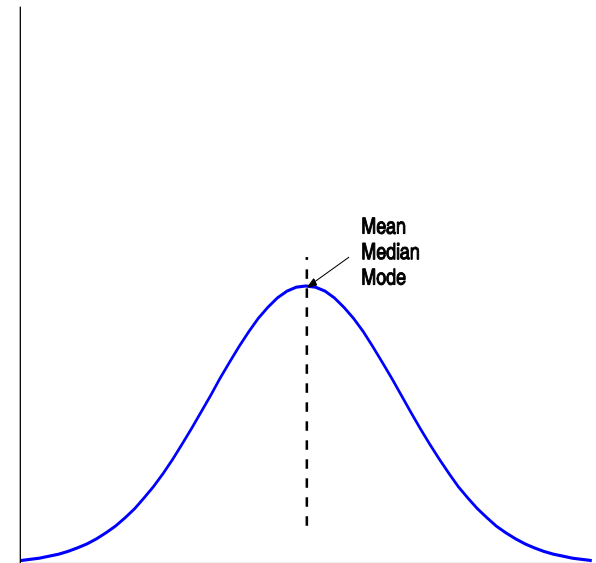
- Value that occurs most frequently in the data
- Unimodal, bimodal, trimodal (i.e., multimodal)
- Empirical formula:

$$mean - mode = 3 \times (mean - median)$$



Symmetric vs. Skewed Data

- Median, mean and mode of symmetric, positively and negatively skewed data





Measuring the Dispersion of Data

- **Quartiles, outliers and boxplots**
 - **Quartiles:** Q_1 (25th percentile), Q_3 (75th percentile)
 - **Inter-quartile range:** $IQR = Q_3 - Q_1$
 - **Five number summary:** min, Q_1 , M, Q_3 , max
 - **Boxplot:** ends of the box are the quartiles, median is marked, whiskers, and plot outlier individually
 - **Outlier:** usually, a value higher/lower than $1.5 \times IQR$
- **Variance and standard deviation (*sample: s, population: σ*)**
 - **Variance:** (algebraic, scalable computation)

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^n x_i^2 - \mu^2$$

- **Standard deviation** s (or σ) is the square root of variance s^2 (or σ^2)



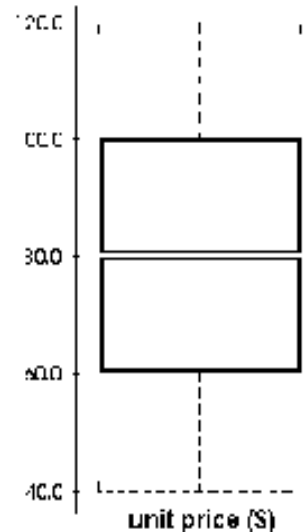
Boxplot Analysis

- **Five-number summary** of a distribution:

Minimum, Q1, M, Q3, Maximum

- **Boxplot**

- Data is represented with a box
- The ends of the box are at the first and third quartiles, i.e., the height of the box is IRQ (i.e., interquartile range)
- The median is marked by a line within the box
- Whiskers: two lines outside the box extend to Minimum and Maximum

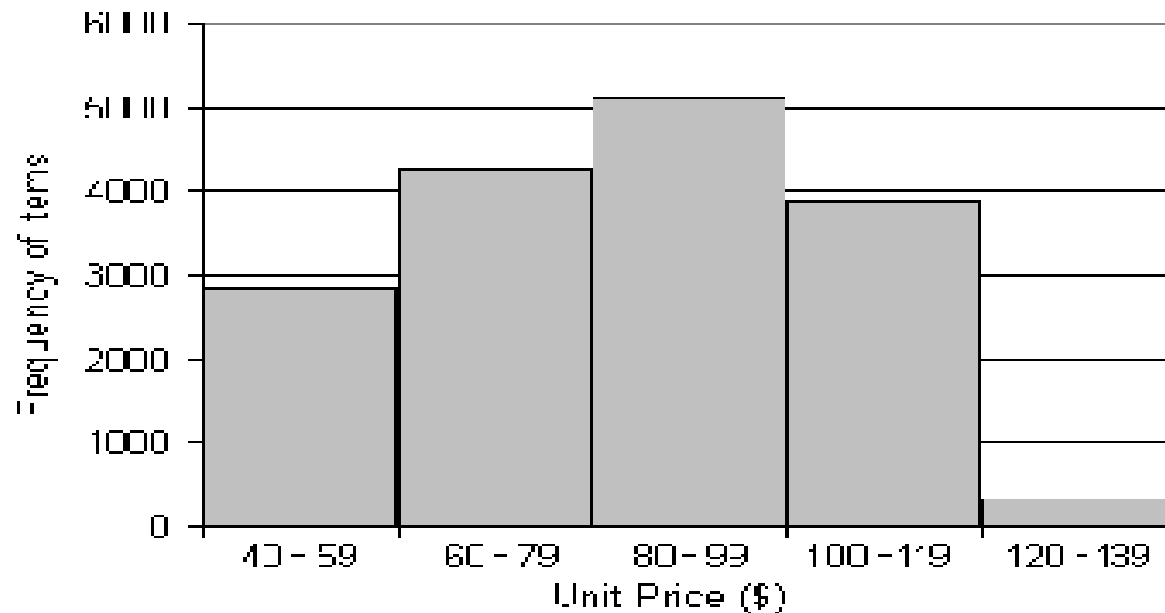




Histogram Analysis

Graph displays of basic statistical class descriptions

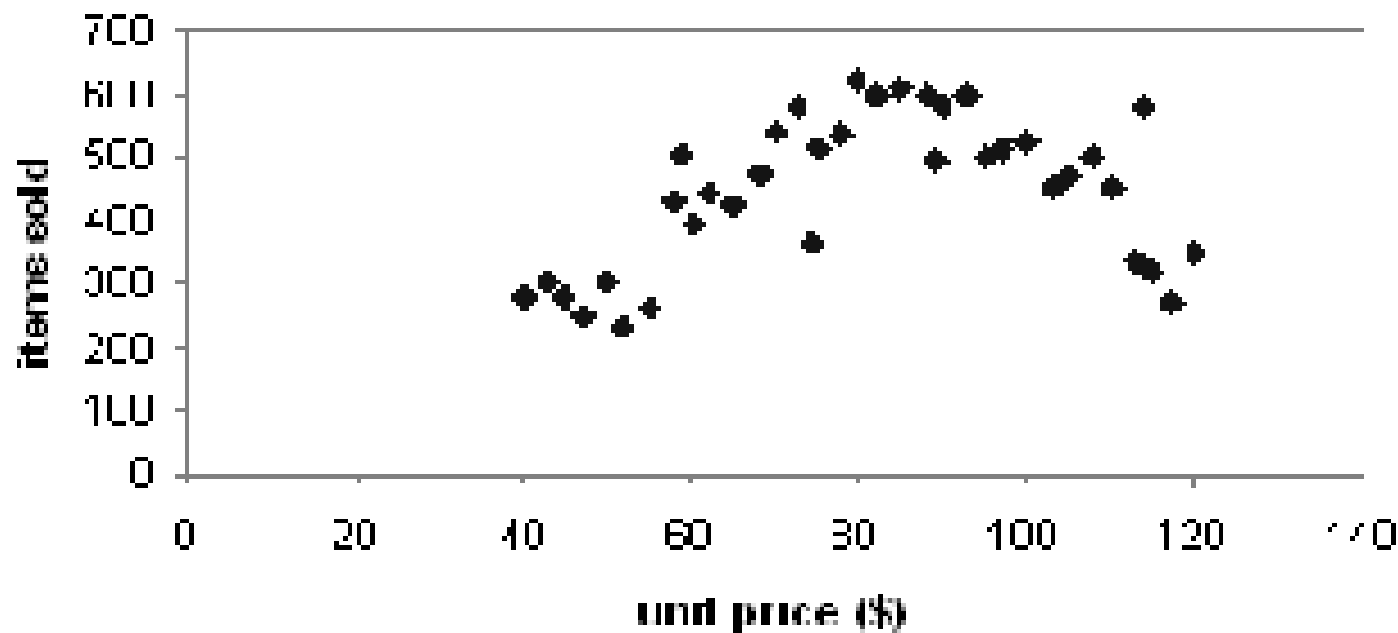
- Frequency histograms
 - A univariate graphical method
 - Consists of a set of rectangles that reflect the counts or frequencies of the classes present in the given data





Scatter plot

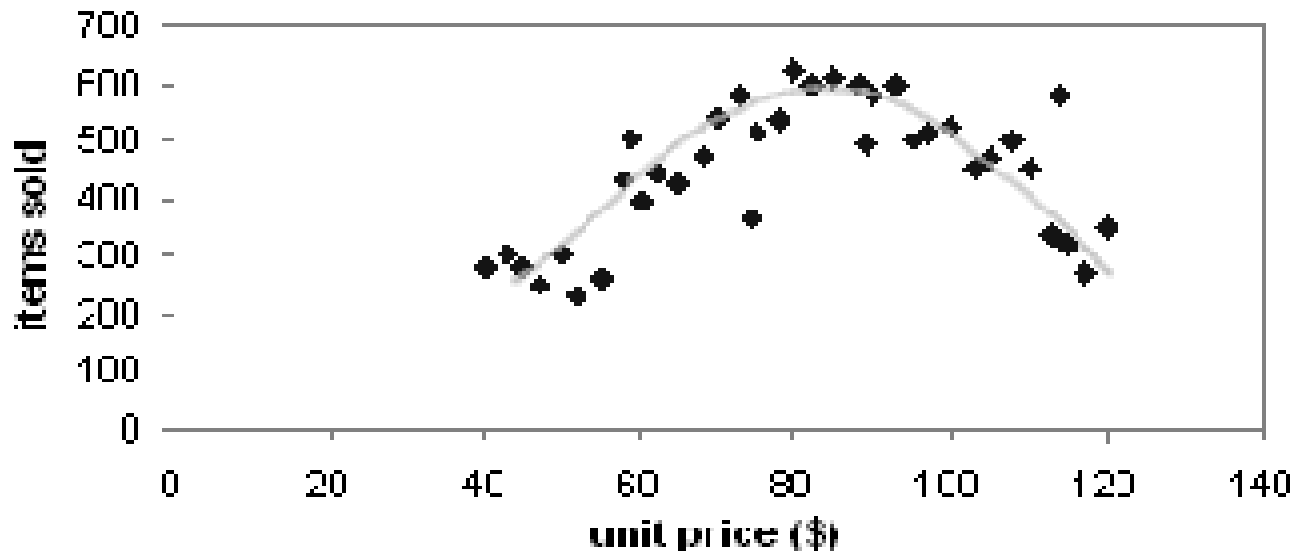
- Provides a first look at bivariate data to see clusters of points, outliers, etc.
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane





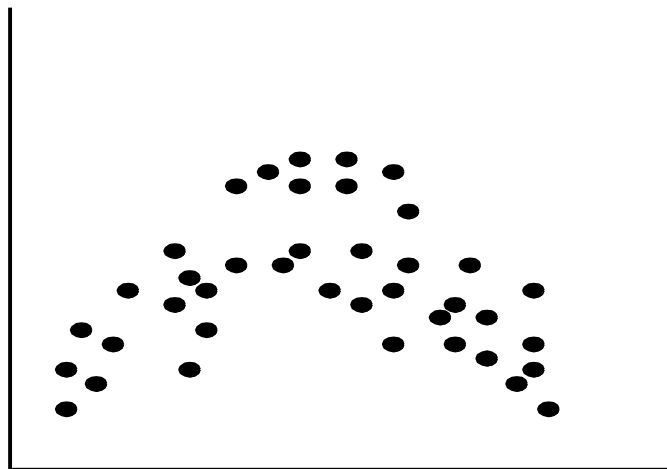
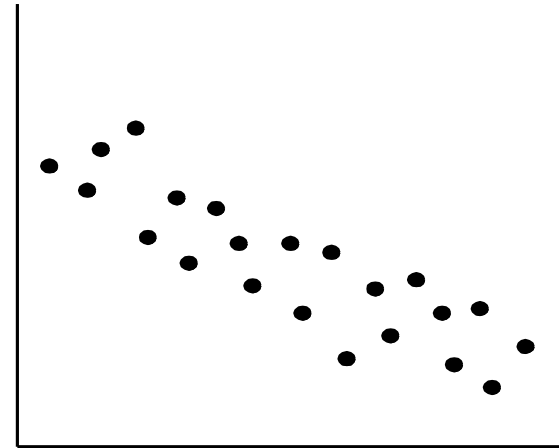
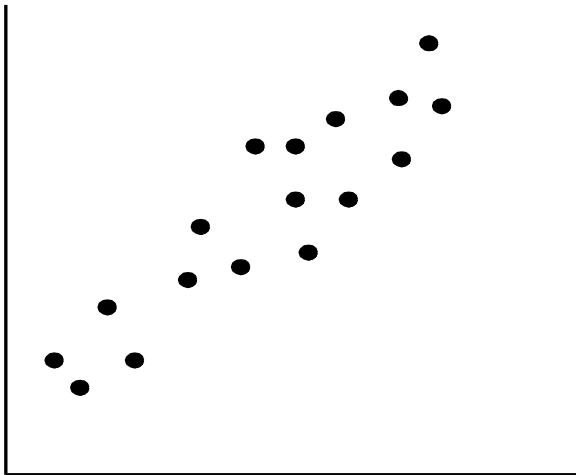
Loess Curve

- Adds a smooth curve to a scatter plot in order to provide better perception of the pattern of dependence
- Loess curve is fitted by setting two parameters: a smoothing parameter, and the degree of the polynomials that are fitted by the regression



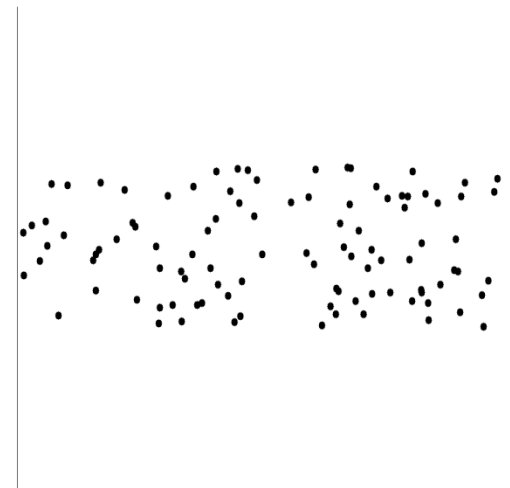
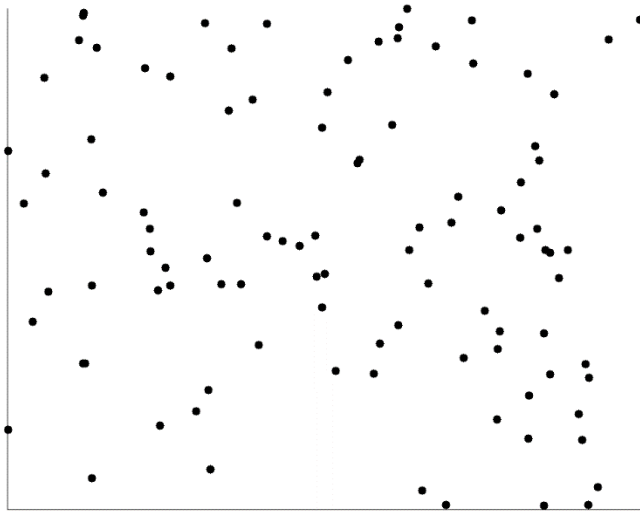
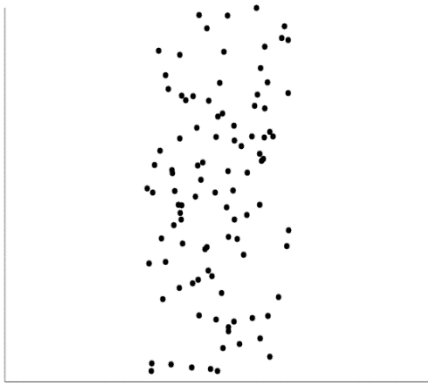


Positively and Negatively Correlated Data





Not Correlated Data





Data Mining vs. Machine Learning

Basic for comparison	Data mining	Machine learning
Meaning	It involves extracting useful knowledge from a large amount of data.	It introduces new algorithm from data as well as past experience.
History	Introduced in 1930 it was initially called knowledge discovery in databases.	It was introduced in 1959.
Responsibility	Data mining is used to examine patterns in existing data. This can then be used to set rules.	Machine learning teaches the computer to learn and understand the given rules.
Nature	It involves human involvement and intervention.	It is automated, once designed it is self-implementing and no or very little human effort is required.